

MagMat: Physics-Grounded Screening & Discovery

Architecture and Methodology Guide for Sustainable Permanent Magnets

Sachin Poudel

Open-Source Platform Guide

Executive Summary

Permanent magnets power electric vehicle traction motors and direct-drive wind generators. Today's highest-performance magnets depend heavily on critical rare-earth elements (Nd, Dy, Tb), whose mining and processing carry severe supply risks and environmental costs.

This framework establishes a physics-grounded screening and discovery methodology for sustainable, rare-earth-free permanent magnets. It unites three complementary capabilities:

1. **Analytical Condensed-Matter Ceilings:** Four classical physical laws (Slater-Pauling, Stoner exchange, Kronmüller micromagnetics, and Maxwell limits) providing parameter-free theoretical upper bounds.
2. **Trained Machine Learning Pipeline:** Multi-task regression and classification ensembles predicting realistic experimental properties (T_C , T_N , ground-state phase, M_s , H_c , K_1 , $(BH)_{\max}$) accompanied by non-parametric conformal uncertainty intervals.
3. **3D Atomistic Foundation AI:** Crystal diffusion (DAO) and equivariant message passing (MACE) resolving polymorph degeneracy and addressing the 89% microstructural defect gap.

1 Methodological Architecture

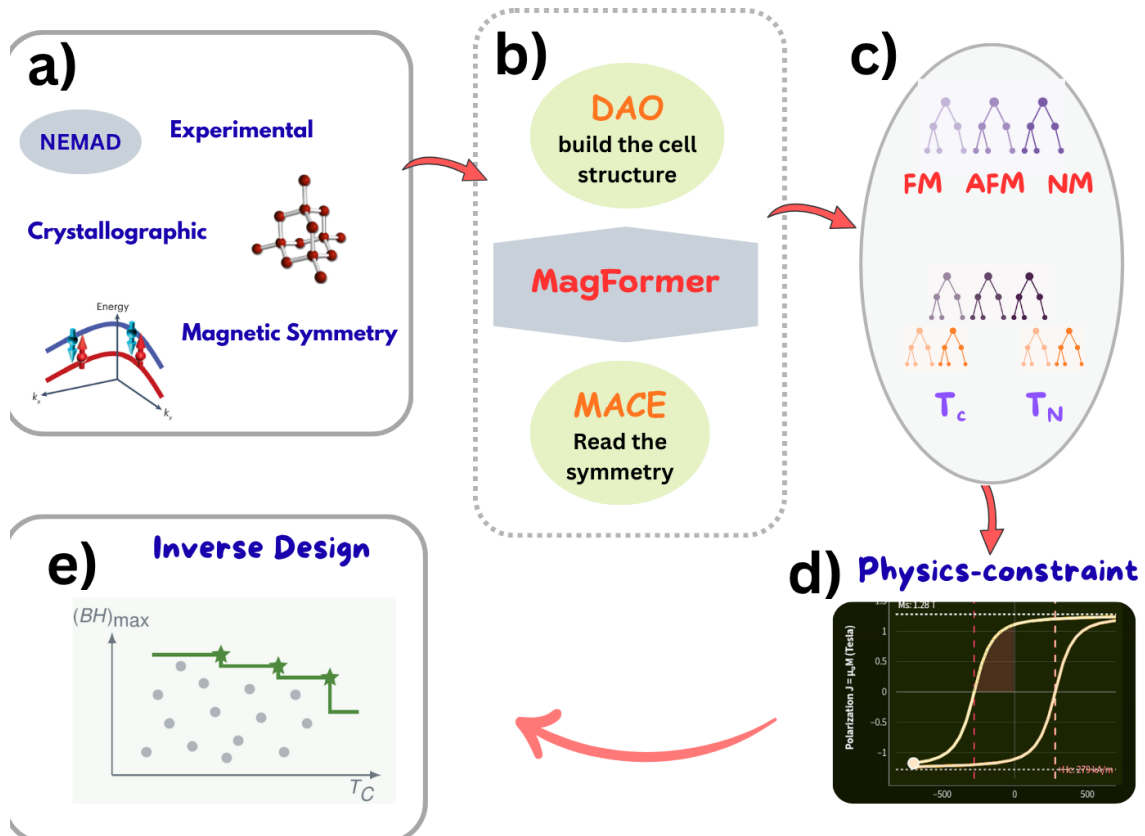


Figure 1: **Multi-Scale Integrated Architecture.** (a) Experimental datasets from NEMAD and literature with crystallographic and magnetic symmetry. (b) 3D foundation AI: DAO crystal diffusion (cell structure and E_{hull} stability), MagFormer compositional transformer, and MACE equivariant symmetry. (c) Trained machine learning models: multi-model classifier for ground-state phase (FM, AFM, NM) and dynamic regression ensembles for transition temperatures (T_C , T_N). (d) Physics-constraint benchmarking against analytical condensed-matter ceilings (Kronmüller micromagnetics, Slater-Pauling, Maxwell bound). (e) Multi-objective inverse Pareto discovery ($(BH)_{\max}$ vs T_C).

The architecture ensures that every composition proposed by machine learning is strictly benchmarked against physical bounds before experimental synthesis or intensive DFT simulations.

2 The Four Analytical Condensed-Matter Ceilings

We implement four classical physical rules that turn chemical stoichiometry into hard performance ceilings without parameter fitting:

#	Physical Bound	Core Question Answered & Governing Physics
1	Slater-Pauling Atomic Ceiling	<i>What is the absolute maximum magnetization?</i> Rigid-band electron filling: majority 3d states fill until valence count $VEC \approx 8.3$ ($\text{Fe}_{70}\text{Co}_{30}$ peak at $2.45 \mu_B/\text{atom}$ / $M_s \leq 245 \text{ emu/g}$). Extra electrons enter minority states and cancel net moment.
2	Stoner Exchange Criterion	<i>Can this chemistry order ferromagnetically?</i> Ferromagnetic ordering requires exchange energy to beat kinetic band broadening cost: $I \cdot D(E_F) > 1$.
3	Kronmüller Micro-magnetics	<i>How hard can an ideal crystal resist demagnetization?</i> Single-crystal coherent rotation sets $H_K = \frac{2K_1}{\mu_0 M_s}$. Real polycrystals reach only $\approx 11\%$ due to defect nucleation.
4	Maxwell Thermodynamic Bound	<i>What is the theoretical maximum energy product?</i> Magnetostatics limits an ideal square loop to $(BH)_{\text{max}} \leq \frac{1}{4} \mu_0 M_s^2$.

Table 1: The four analytical condensed-matter ceilings implemented in the platform.

2.1 Pillar 1: The Slater-Pauling Atomic Band Limit

The Slater-Pauling curve relates the atomic magnetic moment (μ_s , in Bohr magnetons μ_B) to the valence electron concentration (VEC, valence electrons per atom):

$$\bar{\mu}_s = \begin{cases} 2.45 - 0.875(8.3 - N_v) & (5.5 \leq N_v < 8.3) \\ 10.7 - N_v & (N_v \geq 8.3) \end{cases} \quad (1)$$

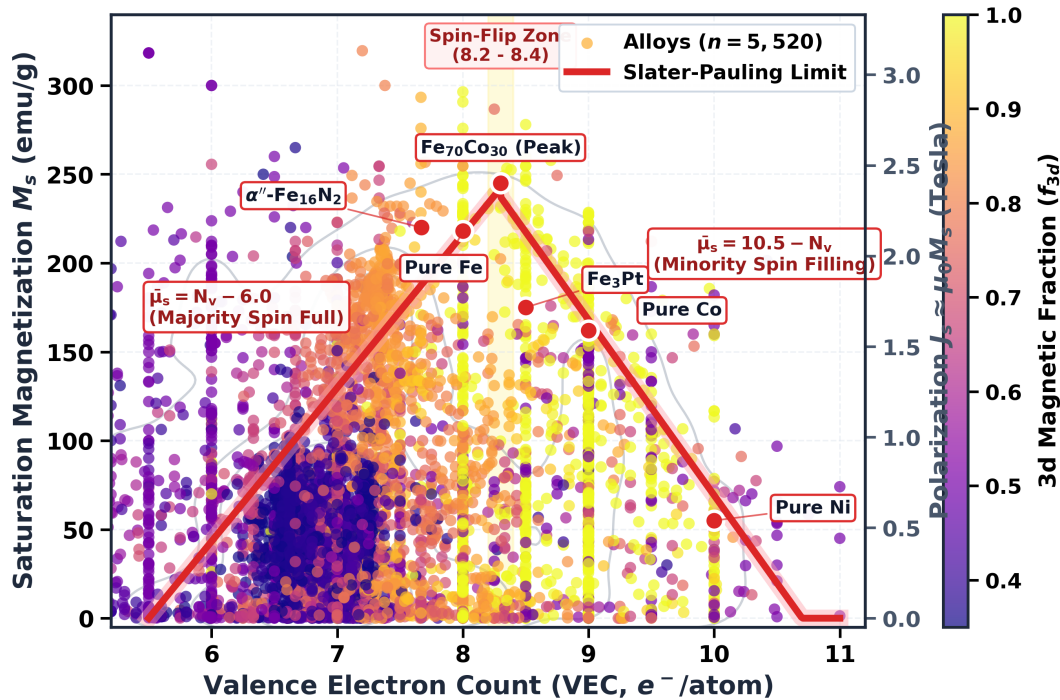


Figure 2: **The Slater-Pauling Saturation Envelope.** Experimental alloy measurements (scatter points) sit beneath the theoretical rigid-band ceiling (red curve), peaking near $VEC \approx 8.25$ in Fe-Co ($2.45 \mu_B$).

Insight on Slater-Pauling

The Slater-Pauling curve is an **upper bound, not a regressor**. Real materials scatter below the ceiling because spin canting, antiferromagnetic sub-lattices, and non-metallic bonding dilute net moments. A record appearing above this line is a diagnostic flag for bad or non-equilibrium experimental data.

2.2 Pillar 2: The 89% Microstructural Defect Gap

While saturation magnetization is an *intrinsic property* determined by composition, coercivity (H_c) is an *extrinsic property* governed by microstructure, grain boundaries, and crystal defects.

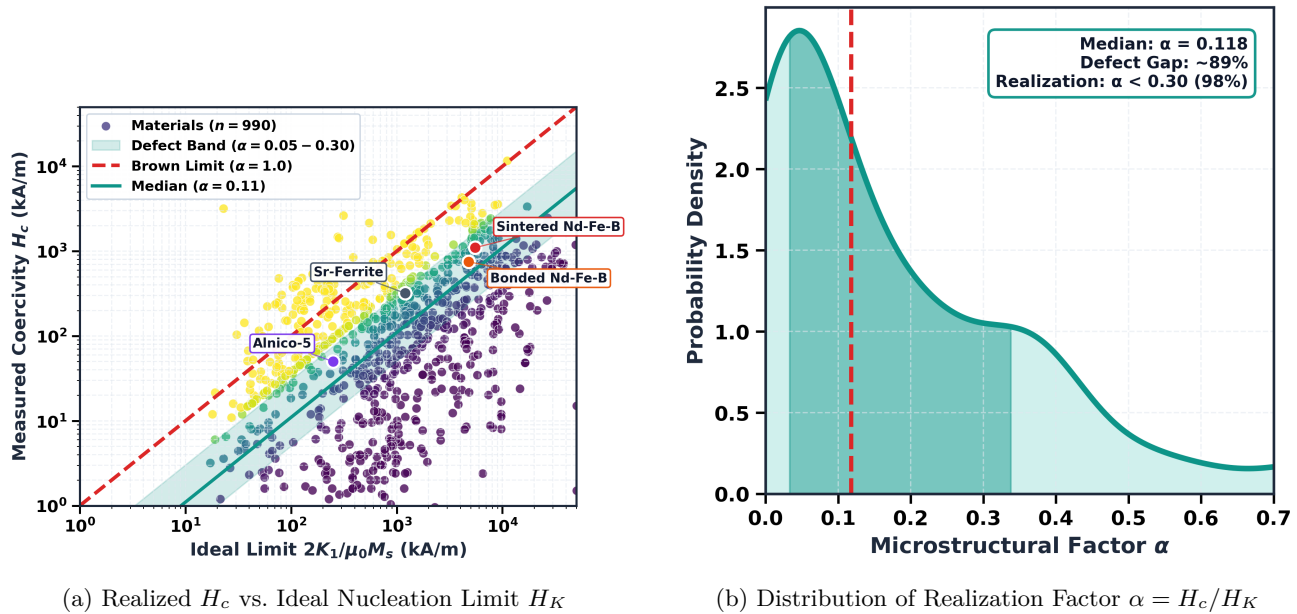


Figure 3: **The Microstructural Coercivity Gap.** (a) Experimental magnets sit one to two orders of magnitude below Brown's ideal limit. (b) The median realization factor across 1,064 records is $\alpha \approx 0.118$, demonstrating an 89% defect gap.

Why 3D Structural AI is Essential

In real sintered magnets, local grain misorientations and defect sites nucleate reverse magnetic domains at fields far below the ideal single-crystal limit ($H_c \approx \alpha \frac{2K_1}{\mu_0 M_s}$). Because this missing 89% headroom lives in geometry rather than chemical formulas, predicting coercivity requires **3D atomistic representations** provided by foundation AI models (DAO and MACE).

3 Harmonized Experimental Data Landscape (56,781 Records)

The dataset integrates 56,781 experimental measurements from NEMAD, Landolt-Börnstein handbooks, and primary literature covering eight core magnetic properties.

3.1 Correlation Drivers of Magnetism

- **Transition-Metal Fraction (f_{3d}):** Strongly drives Curie temperature ($r = 0.45$) and saturation magnetization ($r = 0.31$) through direct $3d-3d$ quantum exchange.
- **Rare-Earth Concentration (f_{RE}):** Governs magnetocrystalline anisotropy (K_1) and coercivity (H_c) via localized $4f$ orbital moments and strong spin-orbit coupling.
- **Valence Electron Count (VEC):** Modulates rigid-band filling, exhibiting a negative correlation with energy product ($(BH)_{\max}$, $r = -0.24$) due to trade-offs between high moment and high coercivity.

4 Trained Machine Learning & 3D Foundation AI

The discovery platform integrates trained multi-task models to deliver experimental property estimates alongside physical bounds:

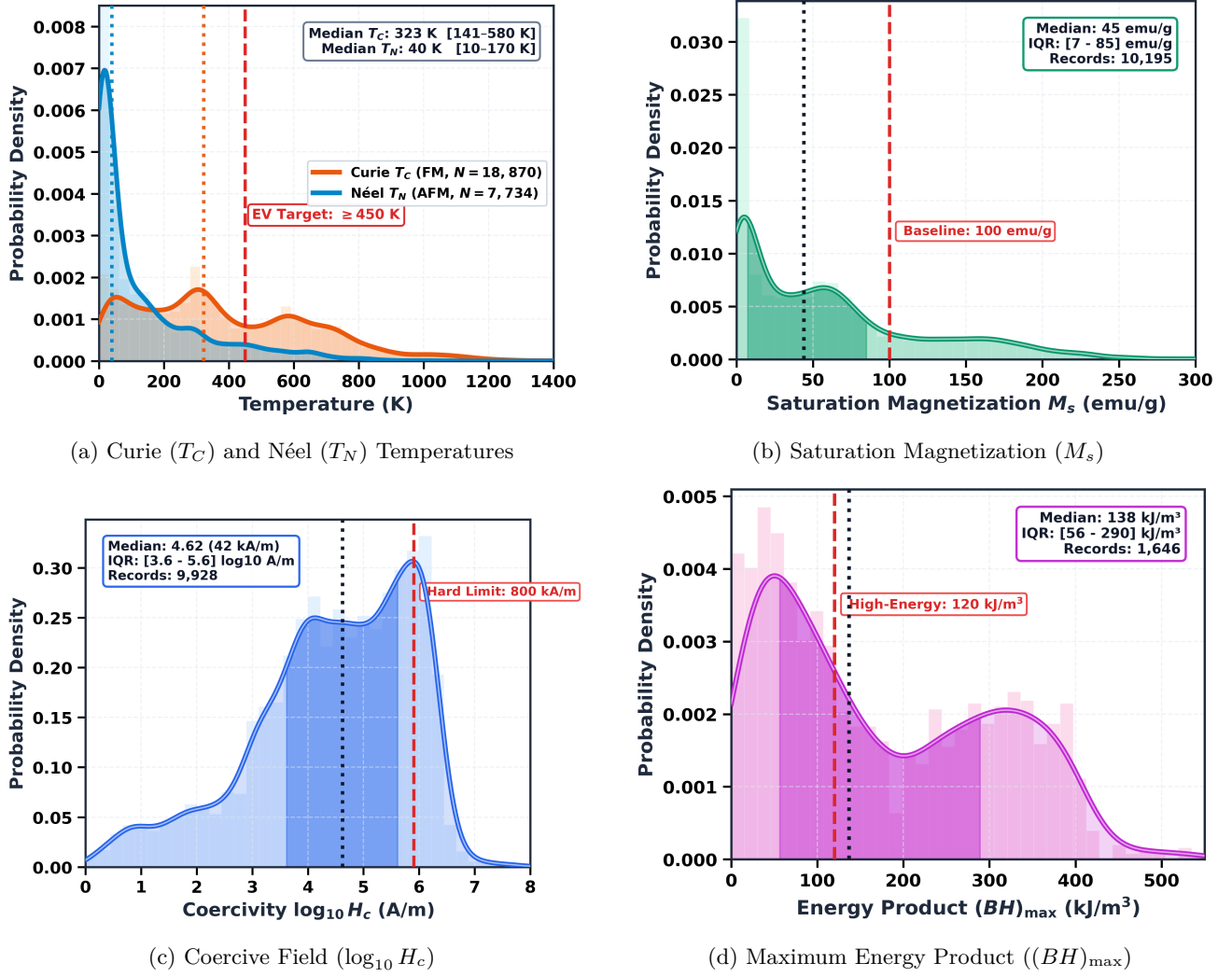


Figure 4: **Empirical Distributions of Core Magnetic Properties.** Showing broad dynamic ranges, multi-modal transition temperatures, and wide coercivity spans across experimental materials.

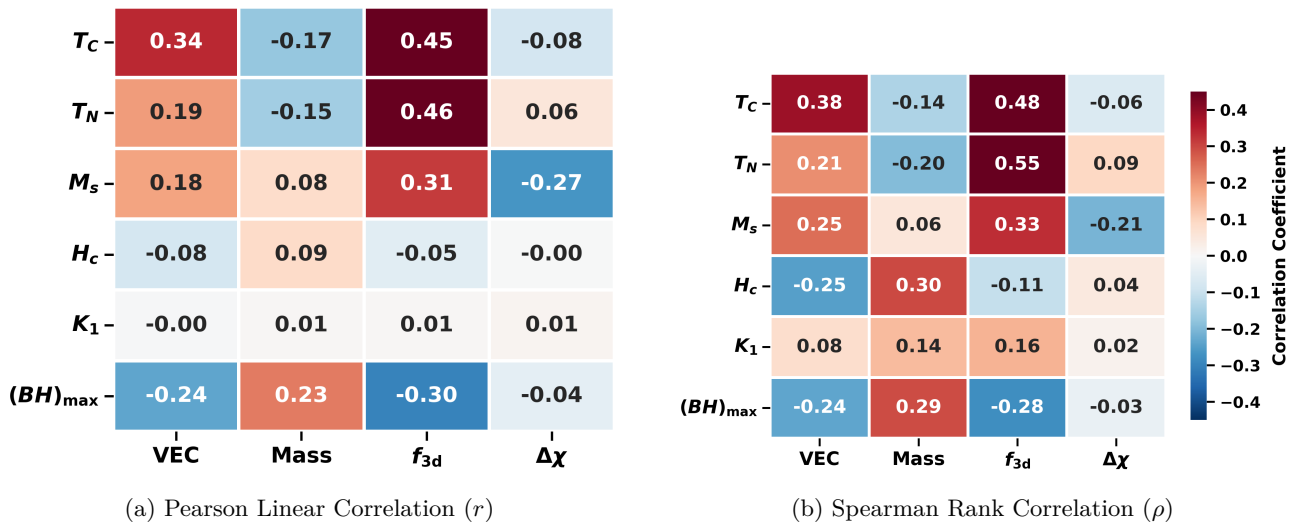


Figure 5: **Correlation Structures.** Linear and rank dependencies between compositional descriptors and macroscopic magnetic performance metrics.

1. Magnetic Phase Classification & Temperature Routing

- **Trained Classifier (phase_clf)**: Multi-model ensemble (LightGBM, XGBoost, CatBoost, ExtraTrees, RF) combined with MagFormer compositional transformer embeddings, achieving **91.9% Accuracy** and **0.908 Macro F_1** .
- **Dynamic Temperature Routing**:
 - If **Ferromagnetic (FM)**: Evaluates the trained T_C regressor ($R^2 = 0.816$, MAE = 69.8 K) and outputs the Curie point.
 - If **Antiferromagnetic (AFM)**: Evaluates the trained T_N regressor ($R^2 = 0.741$, MAE = 48.6 K) and outputs the Néel point, zeroing net moment.
 - If **Non-Magnetic (NM)**: Zeroes ordering temperature and net magnetization.

2. Extrinsic Regressors with Conformal Uncertainty

- **Trained Ensembles**: Multi-task regressors predict M_s , K_1 , H_c , and $(BH)_{\max}$.
- **Non-Parametric Conformal Bounds**: Delivers rigorous 90% confidence intervals without assuming Gaussian residuals, providing trustworthy margins of uncertainty.

3. Pretrained 3D Structural Foundation Models

- **DAO (Diffusion for Atomistic Ordering)**: Generates relaxed 3D crystal unit cells directly from stoichiometry and evaluates thermodynamic stability ($E_{\text{hull}} < 0.1$ eV/atom).
- **MACE (Higher-Order Equivariant Transformer)**: Encodes 3D coordination polyhedra and site symmetries, resolving polymorphs sharing the same stoichiometry (e.g., ferromagnetic τ -MnAl vs. paramagnetic β -MnAl).

5 Inverse Target Discovery & Pareto Screening

Instead of trial-and-error chemistry, the inverse design engine screens candidates against motor application criteria:

$$(BH)_{\max} \geq 120 \text{ kJ/m}^3, \quad T_C \geq 450 \text{ K}, \quad \text{Rare-Earth Free} \quad (2)$$

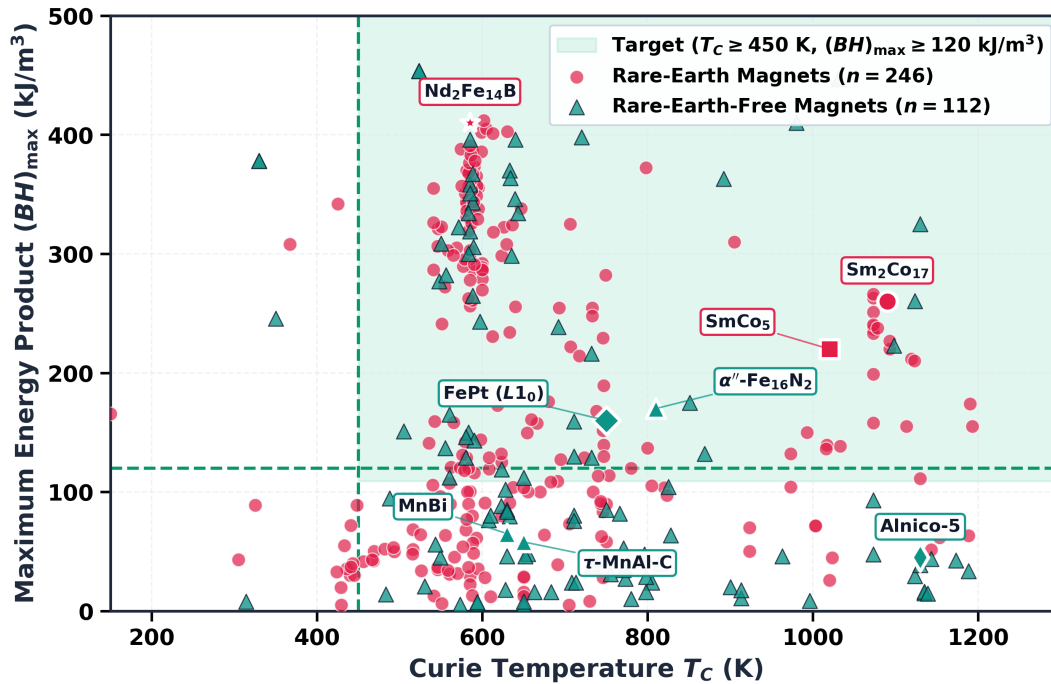


Figure 6: **Inverse Pareto Frontier.** Energy product $(BH)_{\max}$ vs. Curie temperature T_C . The green shaded box represents the target zone for rare-earth-free permanent magnets.

Promising Rare-Earth-Free Candidates on the Pareto Frontier:

1. **FeNi (L1₀ "Tetrataenite")**: Predicted $(BH)_{\max} \approx 335 \text{ kJ/m}^3$ and $T_C \approx 820 \text{ K}$. Highly promising, limited primarily by low-temperature atomic ordering kinetics.

2. τ -**MnAl**: Metastable ferromagnetic phase ($T_C \approx 650$ K, $(BH)_{\max} \approx 60\text{--}80$ kJ/m³), free of critical elements.
3. **Fe₁₆N₂**: Giant saturation magnetization compound with high theoretical energy product headroom.

6 Interactive Exploration via the Web Platform

The entire workflow is accessible interactively via Streamlit:

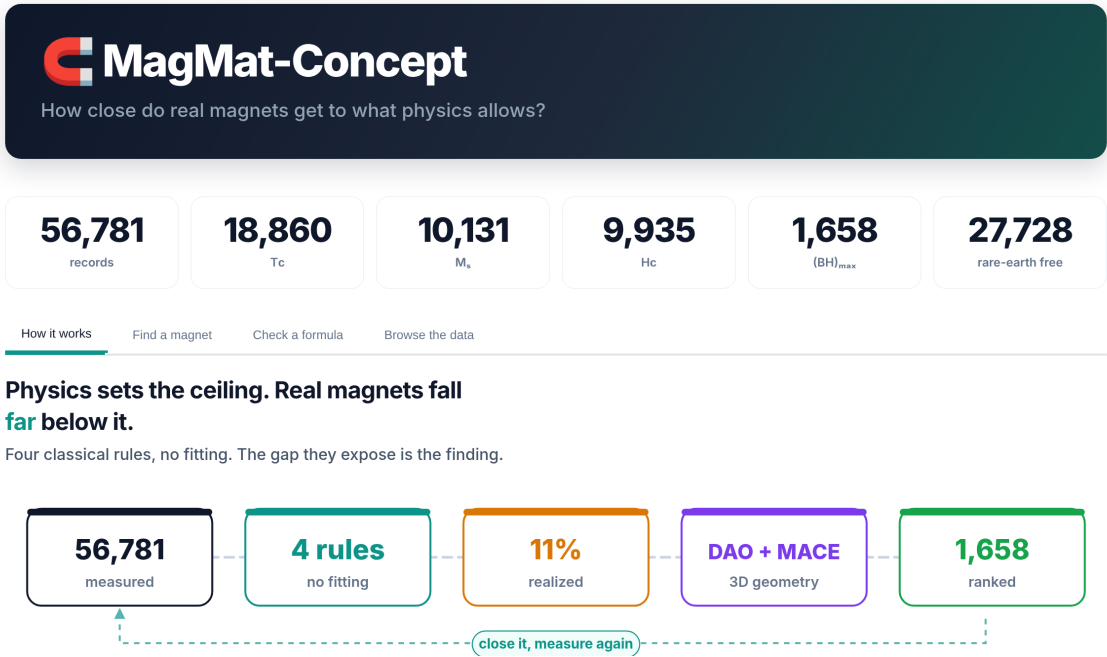


Figure 7: **The Interactive Discovery Platform.** Providing real-time property prediction, 3D unit cell rendering, physical bound calculation, and Pareto candidate exploration.

Launch Command: `streamlit run app/app.py`

7 Summary: Physics vs. Machine Learning Roles

Framework Layer	What It Provides	Role in Materials Discovery
Analytical Ceilings	Slater-Pauling, Stoner, Kronmüller (H_K), and Maxwell limits (parameter-free physical laws).	Serves as an unbending theoretical guardrail; discards unphysical proposals and calculates theoretical headroom.
Trained ML Models	Predictions for Phase (FM/AFM/NM), T_C , T_N , M_s , K_1 , H_c , and $(BH)_{\max}$ with 90% conformal bounds.	Predicts realistic experimental performance and polycrystal properties on harmonized datasets.
3D Foundation AI	DAO crystal diffusion (E_{hull}) and MACE equivariant symmetry representations.	Supplies the spatial and symmetry descriptors needed to bridge the 89% microstructural coercivity defect gap.
Inverse Engine	Multi-objective Pareto optimization against motor thresholds ($(BH)_{\max} \geq 120$ kJ/m ³ , $T_C \geq 450$ K).	Prioritizes high-value, sustainable candidate phases for laboratory synthesis.

Table 2: Synergy between physical ceilings, machine learning, and foundation models.